# RaastaCalm XR

MUHAMMAD OMAR ZAHID L1F22BSCS0028

MUHAMMAD ABDULLAH L1F22BSCS0032

### The Problem:

Young people and professionals are all experiencing the daily commute in silence and are starting to feel the pressure it brings: anxiety, emotional fatigue, and burnout. Anticipating the arrival and constantly “changing modes” at work and home, are not to mention the long travel times and crowded spaces. Mental health apps that require users to sit down, open the app, and be at home, essentially support users only after the accumulated stress has already affected the individual. Soft, micro, on-the-go, and real-world support are what the youth in the early stages of their careers need to address this issue. This is the cycle that so many are stuck in: a fatigued mind leads to a stressed commute, which weighs down on overall focus and well-being.

### The Solution:

RaastaCalm XR (RCX) is the first system to integrate augmented, virtual, and extended reality together to provide a seamless transit-time mental reset system. RCX uses the phone’s gaze-stable AR anchors, via the camera, to provide short, movement-friendly grounding rituals one can do during a commute or while walking through a corridor. For a more thorough reset, users can step into a five-minute VR “Stillness Pod” designed to provide a micro-retreat at calming restorative stations set up throughout workplaces and campuses. The system customizes stress recovery through motion and stress event monitoring. Personal exposure plans, crafted by clinicians, and habit tracking, add to the system’s adaptive stress relief response. RCX’s powerful VR experiences use optional device-only processing and end-to-end encrypted cloud syncing so that user safety and privacy are always maintained.

### The User Journey:

Once a user registers, they complete a brisk onboarding exercise to identify what they want to improve: calmness, focus, energy, or emotional stability. RCX incorporates 2-5 minute AR micro-rituals throughout the day that guide breathing, attention, and grounding with stable, non-disrupting visuals. On the way to campus and the metro stations, users can enter a VR Pod to take a minute or two of immersive time out. After every session, users receive a 30-second reflection, emotional metric trends, and they earn tiny-habit streaks to gamify consistency. Over time users notice that the exact moments in their day that previously felt overwhelming are now filled with improved emotional metrics.

### The Business Impact:

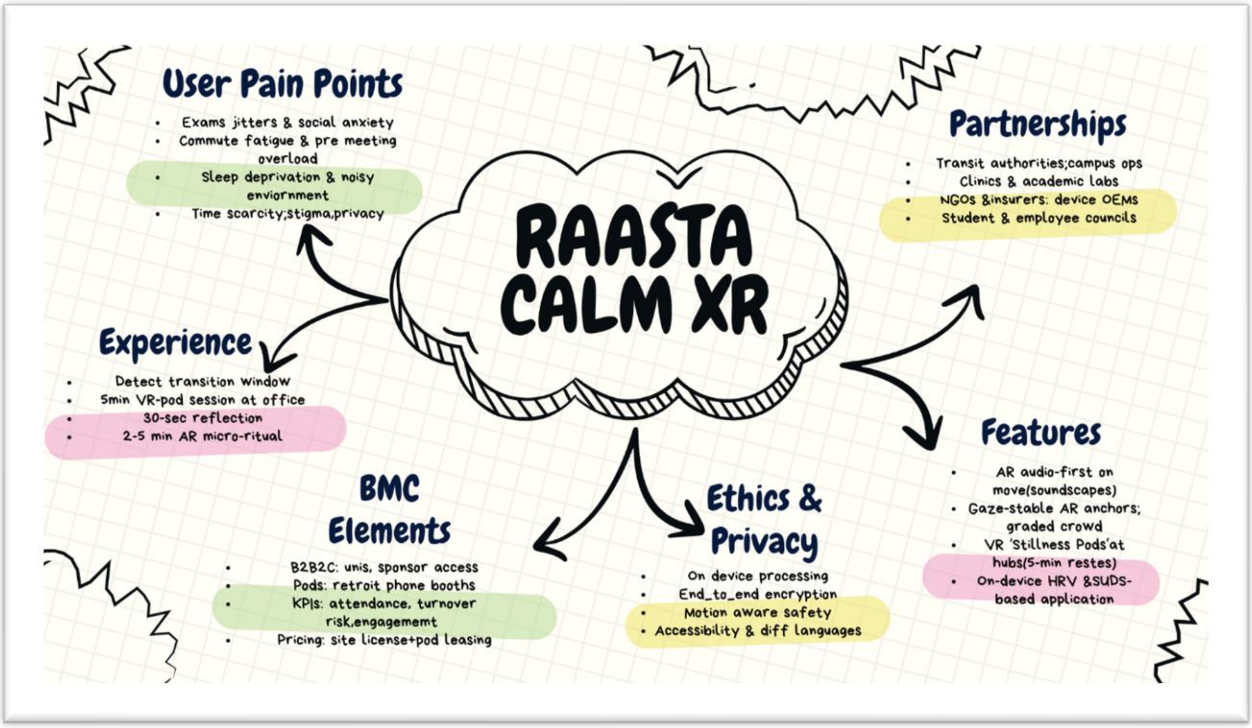
RCX operates under a B2B2C model structure; for wellness infrastructure, universities, co-working spaces, and businesses lease the platform and, if they like, lease the VR pods, which generates continuous income and lessens the user cost. Partners can analyze the metrics of engagement, such as session frequency, micro-break adherence, and pod engagement, to quantify the wellness related productivity impact. Changes in clinically significant indicators like GAD-7 and PHQ-9 shifts help to show measurable improvement of mental health. RCX grows like a network as pods proliferate across campuses and along transit corridors: more sites = less time/emotion for recovery loops = more gainful utilization. The model remains viable due to subscription licensing, pod leasing, and bundled usage data.

### Ethical Lens:

RCX is built with psychological safety at its core. All sensitive processing—stress detection, motion data, AR gaze anchors—stays on-device, with no raw media stored or transmitted. Users control what gets synced, and all personal history is encrypted.

### MindMap:

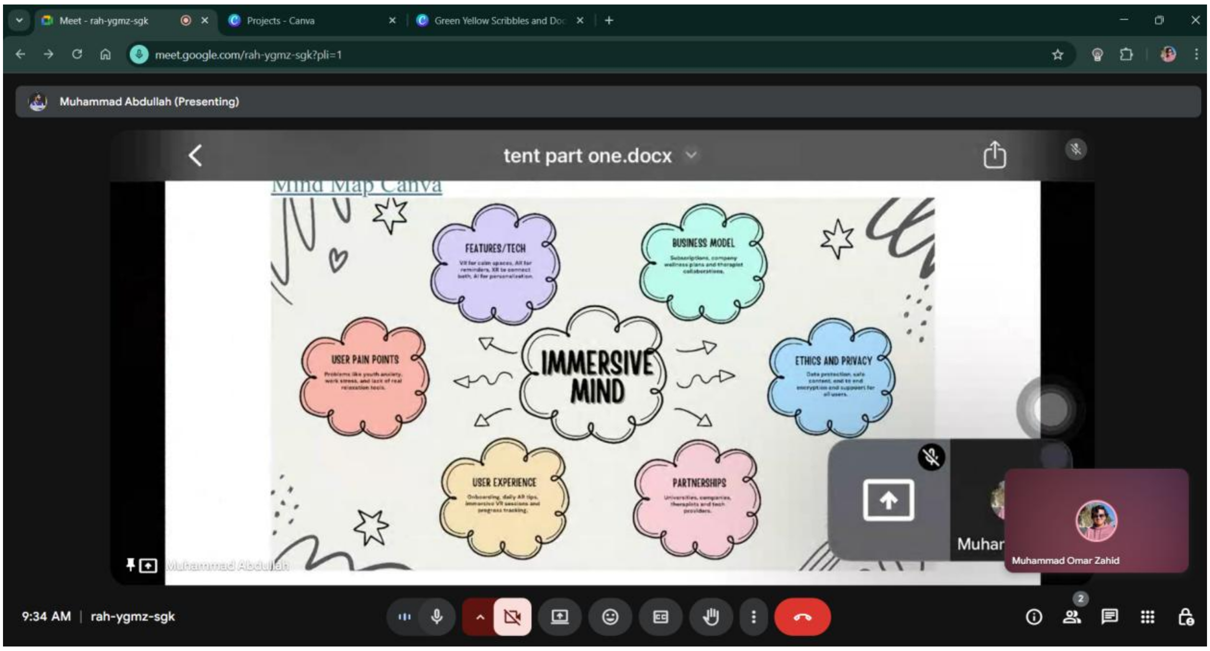
[Link to Mindmap on Canva]




## Daily Work Log:

| Week | Days | Tasks | Status | Comments | Hurdle/Risk | Risk Mitigation |
|---|---|---|---|---|---|---|
| 1 | Monday | Problem research | Completed | Searched different problems | coming up with an idea | stress focusing |
| 1 | Tuesday | Name | Completed | Brainstormed | time consuming | Similarity |
| 1 | Wednesday | Created user personas | Completed | Identified pain points & emotional goals | Personas felt generic | Added transition-based triggers |
| 1 | Thursday | XR technology research | Completed | Confirmed feasibility | Hardware variation | Use low-complexity, gaze-stable AR |

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | Friday | Mind map draft | Completed | First version created & exported | Missing XR integration clarity | Refinement planned for Week 2 |
| 2 | Monday | User journey | Completed | Created complete flow for assignment | Journey too long | Reduced stime to 2–5 min |
| 2 | Tuesday | Write storyline | Completed | Included problem, solution, business, ethics | Storyline too long initially | Compressed |
| 2 | Wednesday | Final MindMap | Completed | Visualized on Canva | Diagram took time | Simplified the visual |
| 2 | Thursday | Ethics & privacy section; refine mind map | Completed | Added on-device processing | Ethical claims needed support | problem |
| 2 | Friday | Finalized | Completed | Made final word file | Missing evidence screenshots | Captured final timestamps |



## Viability & Ethics Analysis:

RaastaCalm XR is viable because it targets a clear and growing need: quick stress-relief for students, workers, and commuters. It fits naturally into places where people already spend idle time, so adoption doesn't require new habits. The model can scale through university and corporate subscriptions, plus optional premium upgrades for individuals. Most features run on existing smartphones, which keeps costs low and makes expansion easier. As more locations join, content can be reused across sites, improving profit margins and making the business sustainable long-term.

Ethics and user protection are built into the design. The system collects only minimal, non-sensitive data such as session length and feature usage, avoiding anything that could identify a user's mental health condition. All information is encrypted, processed securely, and users can opt out of analytics at any point. The experiences are designed as wellness support, not therapy, and avoid medical claims. Content is reviewed with mental-health advisors to ensure it is calming, safe, and not triggering. Safety prompts guide users during VR interactions, and built-in breaks protect against discomfort or overuse. With these measures, RaastaCalm XR balances innovation with privacy, safety, and responsible mental-wellness practice.

**Team Reflection:**

**M. Omar Zahid:**

- Learned how important safety, clarity, and emotional comfort are when designing tech for mental health.
- Saw how immersive environments can calm users and create stronger emotional connection.

**M. Abdullah:**

- Understood how ethics and privacy shape the design of wellness-focused technology.

- Realized that working together brought better ideas and made the final concept stronger.